



# Sadeep Rathnayaka

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## ABOUT ME

A First-Class Honours engineering graduate from the University of Moratuwa, Sri Lanka, and a Full-Stack Engineer with a strong passion for backend systems and database engineering. Combines outstanding academic performance (4x Dean's List, top 0.8% nationally) with specialized international research in robotics and enterprise-level backend software engineering experience. A proactive, rapid learner dedicated to building clean, maintainable codebases, and solving high-impact technical problems.

## EDUCATION AND TRAINING

01/06/2021 – 01/06/2025  
**B.SC. (HONS) IN ENGINEERING (ELECTRONIC AND TELECOMMUNICATION ENGINEERING)** University of Moratuwa, Sri Lanka

- Graduated with **First-Class Honours (GPA: 3.84/4.00)** and **4x Dean's List** distinctions (Semesters 1, 4, 6, 8).
- Completed studies in Electronics and Telecommunication Engineering, specializing in **Software Development and Robotics**.
- Mastered OOP, Data Structures & Algorithms (DSA), Backend Development, and Relational Databases.

Final grade 3.84/4.00 | Level in EQF EQF level 6

01/04/2017 – 01/08/2019  
**G.C.E ADVANCED LEVEL (PHYSICAL SCIENCE STREAM)** Nalanda College, Sri Lanka

- Achieved an Island Rank of **246 out of ~30,000 students** (Top 0.8% nationally) and a District Rank of **80** in Colombo District out of **~4,000 students** (Top 2.0% district-wide).
- Secured a competitive **Z-Score of 2.4172 / 3.00** in the Physical Science stream, securing direct admission to the country's premier engineering university based on district merit.

Final grade Z Score 2.4176 / 3.00 | Level in EQF EQF level 4

## SKILLS

### Programming Languages & Frameworks

Java | Spring Boot | C# | .NET | JavaScript | Angular

### Architecture & Messaging

Microservice, Monolithic | REST-APIs | gRPC | Kafka

### Databases & Tools

SQL | MySQL | Oracle Relational Database | PostgreSQL | Vector Databases | Version Control (Git)

## WORK EXPERIENCE

**SOFTWARE ENGINEER – DIRECTFN PVT LTD.** – 04/05/2026 – Current – COLOMBO, SRI LANKA

- **Backend Development:** Developed enterprise backend business logic using Java by adhering to a standard layered architecture to ensure decoupled, highly maintainable, and structured application features.
- **Database Management:** Managed data schemas in Oracle Database and optimized SQL queries to improve system performance.
- **Frontend Integration:** Built user interfaces using Angular, ensuring the data flow between backend APIs and the UI.
- **Problem Analysis & Debugging:** Investigated technical issues, troubleshoot bugs, and resolved code bottlenecks across both the frontend and backend.
- **Full-Stack Implementation:** Responsible for delivering features end-to-end, covering data table design, API creation, and frontend integration.

**SOFTWARE ENGINEER – IFS R&D INTERNATIONAL PVT LTD.** – 01/07/2025 – 30/04/2026 – COLOMBO, SRI LANKA

- **Architectural Refactoring:** Migrated legacy C# codebases into loosely coupled extensions by implementing interface-driven designs to separate business logic from data layers.
- **Root Cause Analysis (RCA):** Conducted root cause analysis on application errors to identify core system vulnerabilities and implement fixes.
- **Workflow Automation:** Built an automation tool using N8N and Vector Databases to automatically generate unit tests for the migrated codebases.
- **Version Control:** Followed Git best practices to maintain a clean code history and support reliable branch management within the team.

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## ROBOTICS RESEARCH INTERN – SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN – 27/11/2023 – 27/05/2024 – SINGAPORE

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- **Robotic Control Development:** Designed and implemented a functional robotic control system for an autonomous hospital cleaning robot.
- **Framework Implementation:** Utilized the Robot Operating System (ROS) framework to integrate hardware components, manage sensor data, and handle real-time robotic controls.
- **Hardware Testing:** Tested motion control algorithms to ensure the robot operated safely within indoor environments.
- **Research Collaboration:** Worked with international researchers to bridge the gap between theory and real-world solutions.

## PROJECTS

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### DISTRIBUTED BANKING PLATFORM (MICROSERVICE ARCHITECTURE)

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- Built user-service, auth-service, and account-service using **Spring Boot** and **PostgreSQL**, with validation for user registration and profile management.
- Implemented the **Saga microservice pattern** for distributed transactions and guaranteed data consistency across databases.
- Used **Kafka** and **gRPC** with **Protocol Buffers** for strongly typed, efficient inter-service communication.
- Containerized services using Docker to guarantee consistent local execution and deployment-ready packaging.

Link <https://github.com/SadeepRathnayaka/Banking-System>

### MICROSERVICE BASED POS & RETAIL MANAGEMENT PLATFORM

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- Developed a scalable **Spring Boot** backend for a Point of Sales(POS) system using microservices.
- Implemented **Role-Based Access Control** and **JWT authentication** at the **API Gateway** to secure service endpoints.
- Integrated **Eureka Server** for dynamic **service discovery** and **load balancing**.
- Implemented **synchronous inter-service communication** using **Spring WebClient** for interactions between services.

Link <https://github.com/SadeepRathnayaka/Point-of-Sales-System-BackEnd>

### Inventory Control System Backend

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- Developed a scalable **Spring Boot** backend for an inventory system with **CRUD** operations.
- Designed the database schemas with multiple **interconnected tables**, ensuring proper relationships and normalization.
- Optimized **custom query methods** for inventory search with pagination.

Link <https://github.com/SadeepRathnayaka/Inventory-Manager-BackEnd>

## LANGUAGE SKILLS

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Mother tongue(s): **SINHALA**

Other language(s): **ENGLISH - PROFESSIONAL WORKING PROFICIENCY**

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*I am committed to delivering secure, scalable software solutions while continuously expanding my technical skills. I thrive in dynamic environments, bringing strong problem-solving and a collaborative spirit to drive team success.*